TreeWatch Vienna: Street-Level Tree Canopy and Vitality Monitoring with Deep Learning

Akhil Pillai (Matr. No. 12402165)
Applied Deep Learning, TU Wien, Winter Semester 2026
Assignment 1: Initiate

***Abstract*—Cities in the European Union must now prove that they do not lose tree canopy. The EU Nature Restoration Law requires no net loss of urban tree canopy cover by 2030 compared to 2024, and an increasing trend afterwards. Today, most trees are still inspected by hand. TreeWatch is a deep learning system that takes an ordinary phone photo of a street or park tree in Vienna and estimates its species, its size (height, trunk circumference, crown diameter), its approximate age and, as a stretch goal, its vitality. The key idea is that labels come almost for free: the City of Vienna publishes an open tree register with the location and attributes of each tree, so a GPS-tagged photo can be matched to its register entry automatically. This report describes the topic, the project type, related work, the planned dataset, the approach, and a work plan with time estimates.**

***Index Terms*—urban trees, multi-task learning, convolutional neural networks, open data, street-level imagery, Nature Restoration Law**

I. Introduction

Trees are a city's cheapest air conditioning. They provide shade, cool the air and reduce heat stress in densely built areas. Vienna documents its trees in a public tree register and keeps planting more. At the same time, hotter and drier summers put these trees under stress, and many young trees struggle in their first years.

Since 2024 there is also a legal reason to care. The EU Nature Restoration Law [1] (Regulation (EU) 2024/1991, Article 8) requires member states to ensure *no net loss* of urban green space and urban tree canopy cover by 2030 compared to 2024, followed by an increasing trend. Progress has to be monitored and reported. In practice, this means every city needs a reliable and affordable way to know how many trees it has, how big their crowns are, and whether they are healthy.

The problem in simple terms: tree inspections are slow and manual. An inspector walks to each tree, measures it, looks at the crown and writes the result into a database.

The idea: let a neural network do the routine part. A person (or later a vehicle) takes a photo, and the model fills in the tree's "passport". Inspectors then only visit the trees the model flags as suspicious.

II. Topic and Project Type

Topic: computer vision for urban forestry, specifically per-tree attribute estimation from street-level photos, with vitality assessment as a stretch goal.

Project type: *Bring your own data*. I will collect a new dataset of geo-tagged phone photos of Vienna trees and link it to the city's open tree register. I will use existing, well-established methods (pretrained convolutional networks and a segmentation network), adapt them to this new data and establish a baseline for others to compare against. I intend to publish the anonymised dataset at the end of the course. A secondary goal is a small "beat the classics" comparison against simple non-deep-learning baselines (Section V-B).

III. Related Work

Street-level tree catalogues. Branson et al. [2] detected street trees in Google Maps aerial and street-view images and classified them into 40 species, using a public tree inventory of Pasadena for supervision. They detected more than 70% of street trees and assigned the correct species to more than 80%. This shows that street imagery plus an existing tree inventory is enough to train useful models, which is exactly the idea TreeWatch builds on.

Benchmarks and domain shift. The Auto Arborist dataset [3] is a large benchmark with street-level and aerial images of trees from 23 North American cities. Its main finding is that models trained in one city transfer poorly to another. There is no comparable benchmark for a European city, and Vienna's species mix and street design differ from North America.

Tree health from ground photos. Fatima et al. [4] segmented ground-level photos of ash trees into foliage, wood, ivy and background with a U-Net using a ResNet-50 backbone, reaching 84.3% overall accuracy. From the segmentation they derived health indicators such as defoliation, crown symmetry and tilt, and their defoliation estimates agreed well with human observers ($R^2 = 0.83$). Their code and model weights are public. This gives me both a method and a labelling scheme for vitality.

Gap. The works above either focus on detection and species [2], [3] or on the health of a single species with a DSLR camera [4], and all inventory-based work is from North America. TreeWatch combines (1) a European city's open tree register as an automatic source of labels, (2) joint estimation of species, size, age and vitality from a single phone photo, and (3) the framing as a monitoring tool for the Nature Restoration Law [1].

TABLE I
LABELLING ATTRIBUTES IN THE VIENNA TREE REGISTER (233,102 TREES)

Attribute	Format	Missing
Species	Latin + German name	0%
Trunk circumference	centimetres	1.7%
Height	8 classes (0–5 m, 6–10 m, …, >35 m)	1.7%
Crown diameter	8 classes (0–3 m, 4–6 m, …, >21 m)	1.7%
Planting year	year (1708–2026)	27.9%

IV. DATASET

A. Sources

1) **Baumkataster Wien**[1]: open government data with 233,102 city-managed trees in CSV format (export checked on 26.10.2026). Every record has a GPS position and a species name (675 species in 110 genera; the 20 most common genera cover 90% of all trees). The attributes relevant for labelling are shown in Table I.
2) **Own phone photos:** I will walk through parks and streets in Vienna (e.g. Augarten, Prater, Ringstraße, residential streets) and photograph trees. Every photo stores its GPS position and timestamp automatically.
3) **Mapillary**[2]: openly licensed (CC-BY-SA) street-level images. Older summer images will provide views of the same trees *with leaves*, which my own winter photos cannot show.

B. How labels are created

This is the core trick of the project. For each photo:

1) read the GPS position and compass direction of the photo;
2) find the register trees within a few metres in the viewing direction;
3) if exactly one tree clearly matches, copy its register attributes as labels; if the match is ambiguous, discard the photo or check it by hand.

This gives labels for species, height, trunk circumference, crown diameter and age (from planting year) *without manual annotation*. Two details of the register shape the design: height and crown diameter are stored as size *classes* rather than exact metres, so they are predicted as ordered classes; and the planting year is missing for 28% of trees, so the age loss is only computed where a label exists. Only the **vitality** label has to be created by hand.

For it I will follow the foliage/wood segmentation and defoliation scoring of [4], applied to a subset of summer images.

[1]https://www.data.gv.at/katalog/dataset/stadt-wien_baumkatasterderstadtwien

[2]https://www.mapillary.com

C. Planned size and split

- Target: 1,500–3,000 own photos of 800–1,500 individual trees, covering the 20 most common genera (90% of the register; e.g. *Acer*, *Tilia*, *Fraxinus*, *Aesculus*, *Platanus*), plus Mapillary images for the same trees where available.
- Vitality subset (stretch goal): 300–500 hand-labelled images. Ash (*Fraxinus*) is the third most common genus in Vienna with 18,145 trees, which matches the ash-based scheme of [4].
- Split **by tree and by area**, not by photo. Photos of the same tree never appear in both training and test sets, and one whole district is held out as a test of generalisation to new streets.

D. Ethics and privacy

Photos may contain people, faces or number plates. I will blur them automatically before storing or publishing anything, and I will only publish derived data in line with GDPR. Mapillary images will be used under their licence with attribution.

V. APPROACH

A. Model

A pretrained image backbone (e.g. ResNet-50 or a self-supervised model such as DINOv2) is fine-tuned with several small output "heads", one per task. This is called multi-task learning. The **core** tasks are:

- **Genus head:** classification into the 20 most common genera plus "other" (cross-entropy loss), with species as a finer optional head. In winter this relies mainly on bark texture and crown shape.
- **Trunk circumference head:** regression in centimetres (L1 loss).
- **Height and crown heads:** ordinal classification into the register's 8 size classes each.
- **Age head:** regression of age from planting year, masked where the year is missing.

The task losses are combined as a weighted sum, so no single task dominates training.

Stretch goals, attempted only once the core works:

- **Vitality:** a U-Net segments foliage vs. wood; the share of missing foliage gives a defoliation score, as in [4].
- **Multi-photo fusion:** if several photos of one tree exist (different angles or seasons), their features are combined, first by averaging and later with a small attention or recurrent module.

B. Baselines ("beat the classics")

- **Register-only baseline:** predict size from species and age alone (e.g. average height of a 20-year-old plane tree), with no image. If the photo model cannot beat this, the image adds no value.
- **Classical vision baseline:** hand-crafted features (colour histograms, texture descriptors) with a Random Forest.
- **Pl@ntNet**[3] as an external reference for species.

[3]https://plantnet.org

TABLE II
WORK-BREAKDOWN STRUCTURE WITH TIME ESTIMATES

Task	Hours
Literature research and planning (this report)	8
Dataset collection: register download, photo walks, Mapillary download	25
Data pipeline: GPS matching, cleaning, anonymisation, splits	15
Designing and building the network(s) and baselines	15
Training, fine-tuning and experiments	25
Building the web application	15
Writing the final report	10
Preparing the presentation	6
Core total	**119**
Stretch: manual vitality labelling and segmentation	10
Total incl. stretch goals	**129**

C. Evaluation

Top-1 and top-5 accuracy for genus and species; mean absolute error (MAE) for trunk circumference and age; exact and ± 1-class accuracy for height and crown diameter; and, for the vitality stretch goal, intersection over union (IoU) for segmentation and R^2 against my manual vitality scores. Grad-CAM heat maps will show *where* in the image the model looks, for example at bark texture or crown shape.

VI. Application (Assignment 3)

A small web app: the user uploads or takes a photo, and the app shows the predicted tree passport (genus, size and age, plus vitality where the stretch goal is reached) and places the tree on a map of Vienna. If vitality is available, a second view shows a map of trees flagged as "needs inspection".

VII. Work-Breakdown Structure

Table II lists the planned tasks with time estimates. Deadlines: Assignment 2 on 15.12.2026, Assignment 3 on 12.01.2027.

Timeline. Weeks 1–3 (Nov): data collection and pipeline, before most leaves have fallen. Weeks 4–6: baselines and the first multi-task model. Weeks 7–8 (to 15.12): experiments and Assignment 2. Weeks 9–11 (to 12.01): application, report and presentation.

VIII. Risks and Mitigations

- **Season:** most deciduous trees are bare from mid-November. *Mitigation:* collect photos as early as possible, rely on bark and crown shape for genus and size, and use Mapillary summer images for vitality.
- **GPS error (5–10 m in dense streets):** photos may be matched to the wrong tree. *Mitigation:* use only unambiguous matches and check a random sample by hand to measure the error rate.
- **Subjective vitality labels:** I am not a trained arborist. *Mitigation:* follow the published scheme of [4] and report label uncertainty honestly.
- **Register quality:** height and crown diameter are coarse classes, 28% of trees have no planting year, and measurements may be outdated. *Mitigation:* predict classes instead of metres, mask missing ages, and report the remaining label noise and its effect on the error.

Vitality and multi-photo fusion are deliberately planned as stretch goals: if they fail, the genus, size and age results still form a complete project.

IX. Outlook

Every EU city must monitor its tree canopy for the Nature Restoration Law, and most German-speaking cities publish a similar tree register. If the approach works for Vienna, the same pipeline could be transferred to other cities. Possible users include municipalities, landscaping companies, arborists and property managers.

Declaration of AI Tool Usage

In line with TU Wien guidelines, I declare that I used an AI assistant (Claude) for literature search, for validating the bibliographic details of the cited sources, and for checking the wording of this assignment. All cited references were checked against their primary records. The choice of topic, the data collection, the implementation and the experiments are my own work.

References

[1] European Parliament and Council, "Regulation (EU) 2024/1991 on nature restoration and amending Regulation (EU) 2022/869," *Official Journal of the European Union*, 2024.

[2] S. Branson, J. D. Wegner, D. Hall, N. Lang, K. Schindler, and P. Perona, "From Google Maps to a fine-grained catalog of street trees," *ISPRS J. Photogramm. Remote Sens.*, vol. 135, pp. 13–30, 2018.

[3] S. Beery *et al.*, "The Auto Arborist dataset: A large-scale benchmark for multiview urban forest monitoring under domain shift," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 21294–21307.

[4] K. Fatima, A. Prashar, A. Crowe, P. Brown, and R. Gaulton, "Integrated deep learning framework for automating tree health assessment using ground-based images," *Urban Forestry & Urban Greening*, vol. 118, Art. no. 129345, 2026, doi: 10.1016/j.ufug.2026.129345.